

Table 3. Normative age-related total testosterone reference values in nmol/L.

Age	1	2.5	10	20	30	40	50	60	70	80	90	97.5	99
3	0.0	0.0	0.1	0.2	0.3	0.3	0.4	0.4	0.5	0.6	0.7	0.9	1.0
4	0.0	0.0	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.6	0.7	0.9	1.0
5	0.0	0.0	0.1	0.2	0.3	0.3	0.4	0.4	0.5	0.6	0.7	0.8	0.9
6	0.0	0.0	0.1	0.2	0.2	0.3	0.3	0.4	0.4	0.5	0.6	0.8	0.9
7	0.0	0.0	0.0	0.1	0.2	0.2	0.3	0.3	0.4	0.5	0.6	0.7	0.8
8	0.0	0.0	0.0	0.1	0.1	0.2	0.2	0.3	0.3	0.4	0.5	0.7	0.8
9	0.0	0.0	0.0	0.0	0.1	0.1	0.2	0.3	0.4	0.5	0.6	0.8	0.9
10	0.0	0.0	0.0	0.1	0.2	0.2	0.3	0.4	0.5	0.7	0.9	1.2	1.3
11	0.0	0.0	0.2	0.3	0.5	0.6	0.7	0.9	1.1	1.3	1.6	2.1	2.4
12	0.0	0.3	0.8	1.1	1.3	1.5	1.7	2.0	2.4	2.8	3.4	4.3	4.8
13	0.9	1.4	2.2	2.7	3.1	3.5	3.8	4.4	5.1	5.9	7.0	8.8	9.7
14	2.3	3.0	4.4	5.2	5.9	6.4	6.9	8.0	9.2	10.5	12.4	15.3	16.9
15	3.6	4.6	6.6	7.9	8.8	9.5	10.3	11.8	13.4	15.4	18.0	22.1	24.3
16	4.6	5.9	8.3	9.8	11.0	11.9	12.9	14.7	16.7	19.0	22.2	27.2	29.9
17	5.1	6.6	9.3	11.1	12.3	13.4	14.4	16.4	18.6	21.1	24.6	30.0	32.9
18	5.5	7.0	9.8	11.7	13.0	14.1	15.2	17.2	19.4	22.0	25.6	31.1	34.0
19	5.6	7.2	10.0	11.9	13.2	14.3	15.4	17.4	19.6	22.2	25.7	31.1	34.1
20	5.8	7.3	10.1	11.9	13.2	14.3	15.4	17.4	19.5	21.9	25.4	30.7	33.5
21	5.8	7.3	10.0	11.8	13.1	14.2	15.2	17.1	19.2	21.6	24.9	30.0	32.8
22	5.9	7.3	10.0	11.7	12.9	14.0	15.0	16.8	18.8	21.2	24.4	29.3	32.0
23	5.9	7.3	9.9	11.5	12.8	13.8	14.8	16.6	18.5	20.7	23.9	28.7	31.3
24	5.9	7.2	9.8	11.4	12.6	13.6	14.5	16.3	18.2	20.4	23.4	28.1	30.6
25	5.9	7.2	9.7	11.3	12.4	13.4	14.3	16.0	17.9	20.0	23.0	27.6	30.0
26	5.8	7.2	9.6	11.1	12.3	13.2	14.1	15.8	17.6	19.7	22.6	27.1	29.5
27	5.8	7.1	9.5	11.0	12.1	13.1	14.0	15.6	17.4	19.5	22.3	26.7	29.1
28	5.8	7.1	9.4	10.9	12.0	13.0	13.8	15.5	17.2	19.2	22.1	26.4	28.8
29	5.8	7.0	9.3	10.8	11.9	12.8	13.7	15.3	17.0	19.1	21.8	26.2	28.5
30	5.7	7.0	9.3	10.8	11.8	12.7	13.6	15.2	16.9	18.9	21.7	25.9	28.2
31	5.7	6.9	9.2	10.7	11.7	12.7	13.5	15.1	16.8	18.8	21.5	25.8	28.0
32	5.7	6.9	9.1	10.6	11.7	12.6	13.4	15.0	16.7	18.7	21.4	25.6	27.9
33	5.6	6.8	9.1	10.6	11.6	12.5	13.3	14.9	16.6	18.6	21.3	25.5	27.8
34	5.6	6.8	9.0	10.5	11.5	12.4	13.3	14.9	16.5	18.5	21.2	25.4	27.7
35	5.6	6.8	9.0	10.5	11.5	12.4	13.2	14.8	16.5	18.4	21.2	25.4	27.6
36	5.5	6.7	9.0	10.4	11.5	12.3	13.2	14.7	16.4	18.4	21.1	25.3	27.6
37	5.5	6.7	8.9	10.4	11.4	12.3	13.1	14.7	16.4	18.4	21.1	25.3	27.6

Table 3. Cont.

Age	1	2.5	10	20	30	40	50	60	70	80	90	97.5	99
38	5.4	6.7	8.9	10.3	11.4	12.3	13.1	14.7	16.4	18.3	21.1	25.3	27.6
39	5.4	6.6	8.9	10.3	11.3	12.2	13.1	14.7	16.3	18.3	21.1	25.3	27.6
40	5.4	6.6	8.8	10.3	11.3	12.2	13.0	14.6	16.3	18.3	21.1	25.3	27.6
41	5.3	6.5	8.8	10.2	11.3	12.2	13.0	14.6	16.3	18.3	21.1	25.3	27.7
42	5.3	6.5	8.8	10.2	11.3	12.2	13.0	14.6	16.3	18.3	21.1	25.4	27.7
43	5.3	6.5	8.7	10.2	11.3	12.2	13.0	14.6	16.3	18.3	21.1	25.4	27.8
44	5.2	6.4	8.7	10.2	11.2	12.1	13.0	14.6	16.3	18.4	21.2	25.5	27.8
45	5.2	6.4	8.7	10.2	11.2	12.1	13.0	14.6	16.3	18.4	21.2	25.6	27.9
46	5.1	6.4	8.7	10.1	11.2	12.1	13.0	14.6	16.4	18.4	21.2	25.6	28.0
47	5.1	6.3	8.6	10.1	11.2	12.1	13.0	14.6	16.4	18.4	21.3	25.7	28.1
48	5.1	6.3	8.6	10.1	11.2	12.1	13.0	14.6	16.4	18.5	21.3	25.8	28.2
49	5.0	6.3	8.6	10.1	11.2	12.1	13.0	14.6	16.4	18.5	21.4	25.9	28.3
50	5.0	6.3	8.6	10.1	11.2	12.1	13.0	14.6	16.4	18.5	21.4	25.9	28.4
51	5.0	6.2	8.6	10.1	11.2	12.1	13.0	14.6	16.5	18.6	21.5	26.0	28.5
52	4.9	6.2	8.5	10.1	11.1	12.1	13.0	14.7	16.5	18.6	21.6	26.1	28.6
53	4.9	6.2	8.5	10.0	11.1	12.1	13.0	14.7	16.5	18.6	21.6	26.2	28.7
54	4.9	6.2	8.5	10.0	11.1	12.1	13.0	14.7	16.5	18.7	21.7	26.3	28.8
55	4.8	6.1	8.5	10.0	11.1	12.1	13.0	14.7	16.6	18.7	21.8	26.4	28.9
56	4.8	6.1	8.5	10.0	11.1	12.1	13.0	14.7	16.6	18.8	21.8	26.5	29.0
57	4.8	6.1	8.5	10.0	11.1	12.1	13.0	14.7	16.6	18.8	21.9	26.6	29.2
58	4.8	6.0	8.4	10.0	11.1	12.1	13.0	14.8	16.7	18.9	22.0	26.7	29.3
59	4.7	6.0	8.4	10.0	11.1	12.1	13.0	14.8	16.7	18.9	22.0	26.8	29.4
60	4.7	6.0	8.4	10.0	11.1	12.1	13.0	14.8	16.7	19.0	22.1	26.9	29.5
61	4.7	6.0	8.4	10.0	11.1	12.1	13.0	14.8	16.8	19.0	22.2	27.0	29.7
62	4.6	6.0	8.4	10.0	11.1	12.1	13.0	14.8	16.8	19.1	22.2	27.1	29.8
63	4.6	5.9	8.4	10.0	11.1	12.1	13.0	14.9	16.8	19.1	22.3	27.2	29.9
64	4.6	5.9	8.4	10.0	11.1	12.1	13.0	14.9	16.9	19.2	22.4	27.4	30.0
65	4.6	5.9	8.4	10.0	11.1	12.1	13.0	14.9	16.9	19.2	22.5	27.5	30.2
66	4.5	5.9	8.4	10.0	11.1	12.1	13.0	14.9	16.9	19.3	22.5	27.6	30.3
67	4.5	5.9	8.3	10.0	11.1	12.1	13.0	14.9	17.0	19.3	22.6	27.7	30.4
68	4.5	5.8	8.3	10.0	11.1	12.1	13.1	15.0	17.0	19.4	22.7	27.8	30.5
69	4.5	5.8	8.3	9.9	11.1	12.1	13.1	15.0	17.0	19.4	22.8	27.9	30.7
70	4.4	5.8	8.3	9.9	11.1	12.1	13.1	15.0	17.1	19.5	22.8	28.0	30.8
71	4.4	5.8	8.3	9.9	11.1	12.1	13.1	15.0	17.1	19.5	22.9	28.1	30.9
72	4.4	5.8	8.3	9.9	11.1	12.1	13.1	15.0	17.1	19.6	23.0	28.2	31.1

Table 3. Cont.

Age	1	2.5	10	20	30	40	50	60	70	80	90	97.5	99
73	4.4	5.7	8.3	9.9	11.1	12.1	13.1	15.1	17.2	19.6	23.1	28.3	31.2
74	4.3	5.7	8.3	9.9	11.1	12.1	13.1	15.1	17.2	19.7	23.1	28.5	31.3
75	4.3	5.7	8.3	9.9	11.1	12.2	13.1	15.1	17.2	19.7	23.2	28.6	31.5
76	4.3	5.7	8.3	9.9	11.1	12.2	13.1	15.1	17.3	19.8	23.3	28.7	31.6
77	4.3	5.7	8.3	9.9	11.1	12.2	13.1	15.2	17.3	19.9	23.4	28.8	31.7
78	4.3	5.7	8.2	9.9	11.1	12.2	13.1	15.2	17.4	19.9	23.4	28.9	31.8
79	4.2	5.6	8.2	9.9	11.1	12.2	13.1	15.2	17.4	20.0	23.5	29.0	32.0
80	4.2	5.6	8.2	9.9	11.1	12.2	13.2	15.2	17.4	20.0	23.6	29.1	32.1
81	4.2	5.6	8.2	9.9	11.1	12.2	13.2	15.2	17.5	20.1	23.7	29.2	32.2
82	4.2	5.6	8.2	9.9	11.1	12.2	13.2	15.3	17.5	20.1	23.7	29.3	32.4
83	4.2	5.6	8.2	9.9	11.1	12.2	13.2	15.3	17.5	20.2	23.8	29.4	32.5
84	4.1	5.6	8.2	9.9	11.1	12.2	13.2	15.3	17.6	20.2	23.9	29.5	32.6
85	4.1	5.6	8.2	9.9	11.2	12.2	13.2	15.3	17.6	20.3	24.0	29.7	32.7
86	4.1	5.5	8.2	9.9	11.2	12.2	13.2	15.3	17.6	20.3	24.0	29.8	32.9
87	4.1	5.5	8.2	9.9	11.2	12.2	13.2	15.4	17.7	20.4	24.1	29.9	33.0
88	4.1	5.5	8.2	9.9	11.2	12.2	13.2	15.4	17.7	20.4	24.2	30.0	33.1

Ages are in years; column headings are percentiles.
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